



TECHNICAL DESIGN DOCUMENT

TravelPort

Architecture, Data Design & Integration Guide

VERSION

v1.0.0

BACKEND

.NET 8 Clean Architecture

FRONTEND

React 18 + TypeScript

DATABASE

SQL Server + EF Core 8

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Clean Architecture Overview

Layer responsibilities and dependency rules

1.1 Layer Responsibilities

API (Presentation) — Controllers, Middleware (JWT, exception handling, logging), Action Filters, Program.cs (DI root)

depends on ↓

Application (Business Logic) — AuthService, FlightService, HotelService, BookingService, WalletService, AdminService; DTOs, FluentValidation validators, ICacheService, IUnitOfWork

depends on ↓

Domain (Core) — Entities (User, Flight, Hotel, Booking, Wallet, WalletTransaction, Coupon, RefreshToken), Enums (BookingStatus, BookingType, TravelClass), ValueObjects — ZERO external dependencies

Infrastructure & Persistence implement interfaces defined in ↑

Persistence — TravelPortDbContext, Repositories (Flight, Hotel, Booking, Wallet, Coupon, User, Admin), EF Core Migrations, DataSeeder

Infrastructure — JwtService, CacheService, SmtplibEmailService, RazorpayService, DuffelProvider; BackgroundServices (BookingExpiryWorker, RefreshTokenCleanupWorker)

1.2 Dependency Rule

```
Domain ← Application ← Infrastructure ← API
      ↑           ↑
      Persistence External APIs
```

```
# Arrows point INWARD. Outer layers know about inner layers.
# Inner layers (Domain) know nothing about outer layers.
# This is enforced via project references in .csproj files.
```

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Backend Design

Entities, services, repositories and caching

2.1 Core Domain Entities

ENTITY	KEY FIELDS	NOTES
User	Id, Email, PasswordHash, Role, IsBlocked, WalletId	Wallet created on registration via cascade
Flight	Id, FlightNumber, Origin, Destination, Airline, DepartureTime, ArrivalTime, AvailableSeats, BasePrice	900+ seeded across 42 routes
Hotel	Id, Name, City, StarRating, PricePerNight, Amenities, RoomTypes	60+ across 12 cities
Booking	Id, BookingReference (FL/HT prefix), UserId, FlightId?, HotelId?, FinalAmount, Status, BookingType	Status: Pending → Confirmed / Cancelled
Wallet	Id, UserId, Balance	One wallet per user; balance validated before deduction
WalletTransaction	Id, WalletId, Amount, Type (Credit/Debit), Description, ReferenceId	Immutable audit trail
Coupon	Id, Code, DiscountType (Percentage/Fixed), DiscountValue, MinOrderValue, IsActive, CouponType	Flight-specific or Hotel-specific
RefreshToken	Id, UserId, Token, ExpiresAt, IsRevoked	Cleaned up by background worker every 24h

2.2 Caching Strategy

CACHE KEY PATTERN	TTL	INVALIDATED ON
flights:{Origin}:{Destination}:{Date}:{Pax}:{Class}	5 min	Flight booking (seat count change)
flight:{id}	30 min	Flight booking

CACHE KEY PATTERN	TTL	INVALIDATED ON
<code>hotels:{City}:{In}:{Out}:{Guests}:{Rating}:{Sort}</code>	5 min	Hotel booking
<code>hotel:{id}</code>	30 min	Hotel booking

`ICacheService` wraps `IDistributedCache` with JSON serialization. Default: `AddDistributedMemoryCache()`. Production swap: replace with `AddStackExchangeRedisCache()` in Program.cs — zero service-layer changes.

2.3 Atomic Booking + Wallet

```
// FlightService.BookAsync - simplified flow
1. ValidateFlightExists(flightId)
2. CheckSeatsAvailable(flight, passengerCount)
3. couponDiscount = ApplyCoupon(couponCode, baseAmount) // optional
4. if (useWallet) _wallet.DeductAsync(userId, finalAmount) // NO SaveChanges yet
5. booking = new Booking { ... Status = Pending }
6. _bookings.AddAsync(booking)
7. flight.AvailableSeats -= passengerCount
8. _unitOfWork.SaveChangesAsync() // ← ONE atomic commit: booking + wallet + seats
9. _cache.RemoveAsync("flights:{key}")
10. _email.SendBookingConfirmed(booking)
```

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Frontend Architecture

Feature modules, state management and API integration

3.1 Feature Module Structure

```
frontend/src/
├── api/
│   ├── axios.ts           ← Axios instance + JWT interceptor (auto-refresh)
│   └── endpoints.ts       ← All API URL constants (no hardcoded URLs elsewhere)
├── components/
│   ├── common/           ← Button, Select, Modal, Skeleton, ConfirmDialog
│   └── ui/               ← Design system primitives
├── features/
│   ├── auth/             ← authSlice (Redux), LoginForm, RegisterForm
│   ├── flights/          ← FilterSidebar, FlightCard, FarePopup, TravellerDetails
│   ├── hotels/           ← HotelCard, HotelFilters, RoomSelector
│   ├── bookings/        ← BookingCard, BookingDetailPage, PDFDownload
│   └── admin/            ← AdminPage (4 tabs: Dashboard, Users, Bookings, Coupons)
├── pages/               ← Route-level components (FlightsPage, HotelsPage, etc.)
├── services/            ← flightService, hotelService, bookingService, userService
├── store/               ← Redux store + authSlice
├── types/index.ts       ← All shared TypeScript interfaces
└── routes/              ← AppRouter, PrivateRoute, AdminRoute
```

3.2 JWT Interceptor Pattern

```
// api/axios.ts
api.interceptors.response.use(
  (res) => res,
  async (error) => {
    if (error.response?.status === 401 && !originalRequest._retry) {
      originalRequest._retry = true;
      const token = await refreshAccessToken(); // POST /auth/refresh
      api.defaults.headers.common['Authorization'] = `Bearer ${token}`;
      return api(originalRequest);
    }
    return Promise.reject(error);
  }
);
```

3.3 Key Design Choices

DECISION	CHOICE	REASON
State management	Redux Toolkit	Predictable auth state; DevTools support; minimal boilerplate
Validation	Zod	Runtime type safety matching backend FluentValidation contracts
Styling	Tailwind CSS v3	Utility-first; no CSS-in-JS overhead; purged in production
Build tool	Vite 6	Fast HMR; native ESM; proxy config for /api → backend
HTTP	Axios	Interceptor support for JWT refresh; typed response wrapper
Named exports	All components	Avoids default-export aliasing confusion at import sites

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Database Design

Core tables, relationships and migration strategy

4.1 Entity Relationships

RELATIONSHIP	CARDINALITY	FK
User → Wallet	1 : 1	Wallet.UserId (Cascade delete)
User → RefreshToken	1 : many	RefreshToken.UserId
User → Booking	1 : many	Booking.UserId
Wallet → WalletTransaction	1 : many	WalletTransaction.WalletId
Flight → Booking	1 : many (nullable)	Booking.FlightId
Hotel → Booking	1 : many (nullable)	Booking.HotelId
Hotel → RoomType	1 : many	RoomType.HotelId

4.2 Migration Strategy

EF Core Code-First migrations. Auto-applied on API startup via `db.Database.Migrate()` in `Program.cs`. DataSeeder runs after migration — idempotent checks prevent re-seeding on restart.

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Authentication Flow

JWT + Refresh token rotation

- 1
- Login
- POST /auth/login → validate credentials (BCrypt) → issue JWT (15 min) + RefreshToken (7 days, stored in DB)
- 2
- API Call
- Client sends Authorization: Bearer {access_token} header

Axios interceptor catches 401 → POST /auth/refresh → new access token issued

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Token Rotation

Old refresh token revoked; new pair issued. IsRevoked flag set in DB.

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Cleanup

RefreshTokenCleanupWorker deletes revoked/expired tokens every 24 hours

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External Integrations

RAZORPAY

Payment gateway. Enabled via config flag. Mock fallback in dev returns success.

SMTP / OFFICE365

Transactional email. HTML table layouts for Gmail/Outlook compatibility. TLS 587.

DUFFEL

Real flight search API. Disabled by default — rich DB seed data used instead.